

Exp No: 11

STUDY OF GPIO & BIT ADDRESSING IN AT89C51

AIM:

To study GPIO control and bit-addressable SFR using Embedded C.

SOFTWARE REQUIRED:

- Keil

THEORY

8051 ports are byte and bit addressable. Using sbit, individual bits of a port can be accessed

directly. Ports default to logic HIGH due to internal pull-ups.

PROGRAM:

```
#include <reg51.h>
```

```
sbit LED1 = P1^0;
```

```
sbit LED2 = P1^1;
```

```
void delay()
```

```
{
```

```
    unsigned int i,j;
```

```
    for(i=0;i<200;i++)
```

```
        for(j=0;j<1000;j++);
```

```
}
```

```
void main()
```

```
{
```

```
    while(1)
```

```

{
LED1 = 1;

LED2 = 0;

delay();

LED1 = 0;

LED2 = 1;

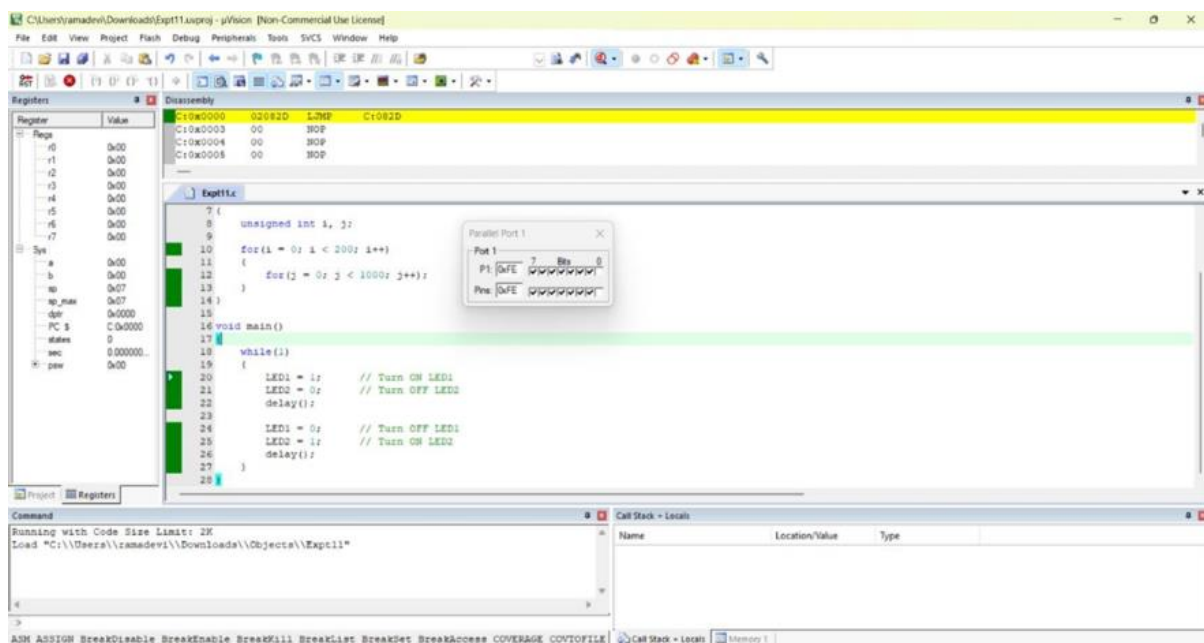
delay();

}

}

```

OUTPUT:



RESULT:

Thus GPIO and bit addressing was studied and in Port1 the pins P1.0 and P1.1 toggled alternately.